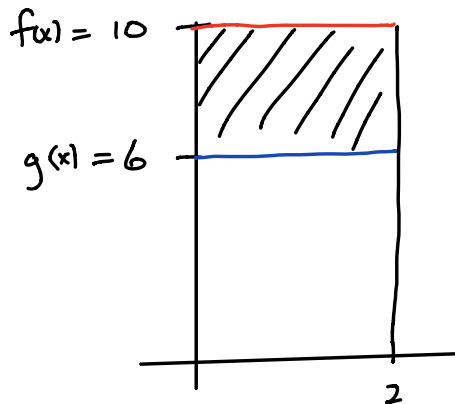


Section 6.1 Area Between Curves

Ex ① Given $f(x) = 10$ and $g(x) = 6$, find the area between $f(x)$ and $g(x)$ on the interval $[0, 2]$.



$$\text{Area under } f(x) = 10 \cdot 2 = 20$$

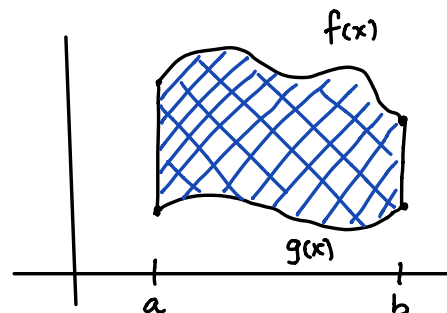
$$\text{Area under } g(x) = 6 \cdot 2 = 12$$

$$\text{Area between } f(x) \text{ and } g(x) \text{ is } 20 - 12 = \textcircled{8}$$

$$\text{In other words, } \int_0^2 (f(x) - g(x)) dx = \textcircled{8}$$

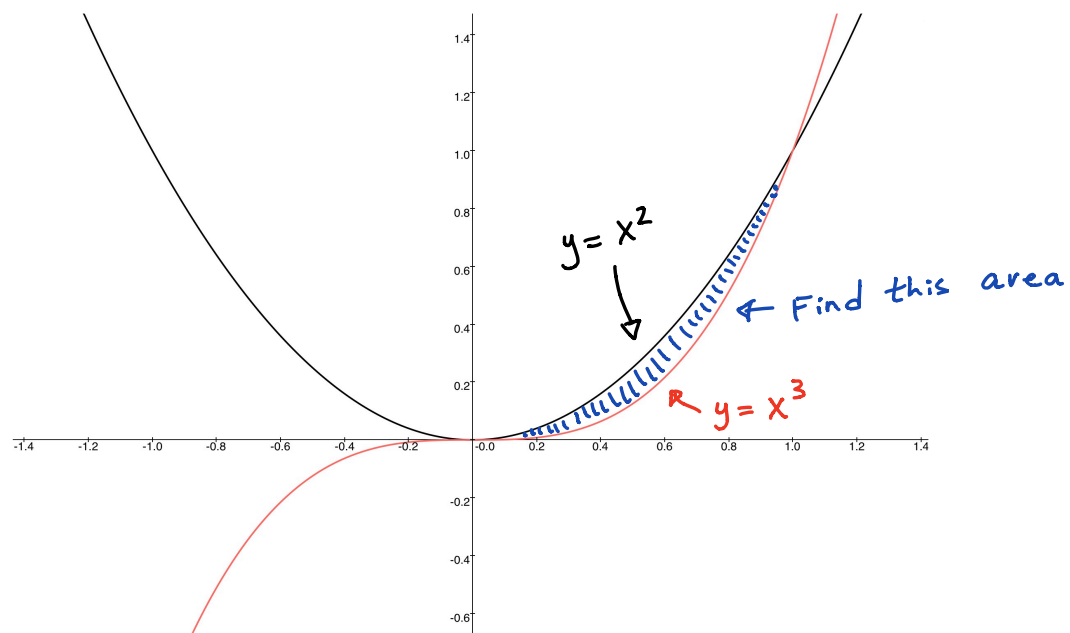
In general, if $f(x) \geq g(x)$ on $[a, b]$ then the area between $f(x)$ and $g(x)$ is

$$\int_a^b \overset{\text{top - bottom}}{(f(x) - g(x))} dx$$



Ex ② Find the finite area between
 $y = x^2$ and $y = x^3$.

First look at the graph to determine
the area asked for.



Now find the intersection points.

$$x^2 = x^3$$

$$0 = x^3 - x^2$$

$$0 = x^2(x-1)$$

$$\begin{array}{cc} \downarrow & \downarrow \\ x=0 & x=1 \end{array}$$

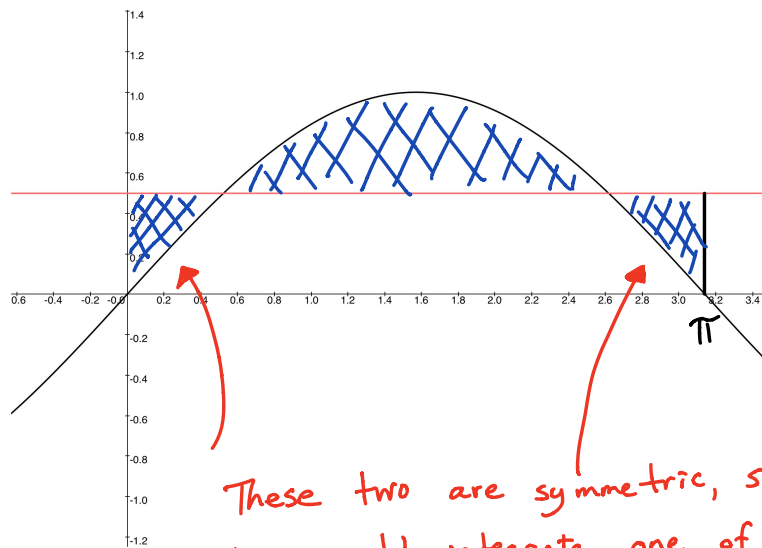
$$\int_0^1 \overset{\text{top} - \text{bottom}}{(x^2 - x^3)} dx = \left(\frac{x^3}{3} - \frac{x^4}{4} \right) \Big|_0^1$$

$$= \left(\frac{1}{3} - \frac{1}{4} \right) - \left(\frac{0}{3} - \frac{0}{4} \right)$$

$$= \frac{1}{12}$$

Ex ③ Find the area between $y = \sin x$ and $y = \frac{1}{2}$ on $[0, \pi]$.

Take a look at the graph.



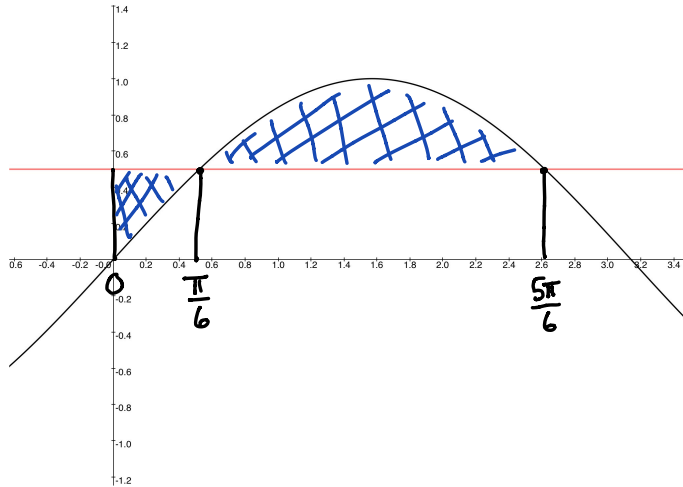
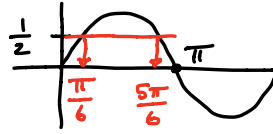
These two are symmetric, so we could integrate one of these and double that answer.

Find the intersection points first.

$$\sin x = \frac{1}{2}$$

$$x = \sin^{-1}\left(\frac{1}{2}\right)$$

$$x = \frac{\pi}{6} \text{ or } x = \frac{5\pi}{6}$$



$$\text{Area} = 2 \cdot \int_0^{\frac{\pi}{6}} \left(\frac{1}{2} - \sin x\right) dx + \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \left(\sin x - \frac{1}{2}\right) dx$$

$$= 2 \left(\frac{1}{2}x + \cos x \right) \Big|_0^{\frac{\pi}{6}} + \left(-\cos x - \frac{1}{2}x \right) \Big|_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$$

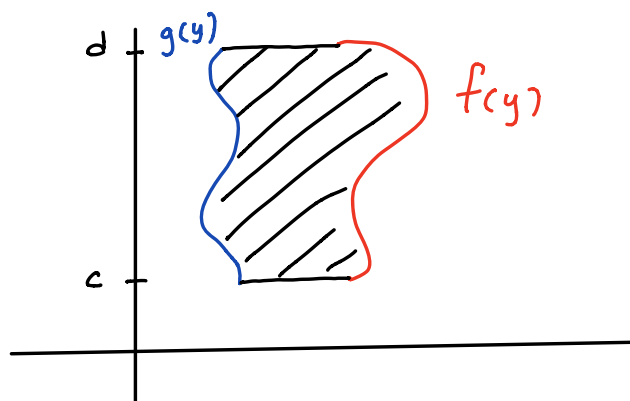
$$= 2 \left(\frac{1}{2} \cdot \frac{\pi}{6} + \cos \frac{\pi}{6} - \left(\frac{1}{2}(0) + \cos(0) \right) \right)$$

$$+ \left(-\cos \frac{5\pi}{6} - \frac{1}{2} \cdot \frac{5\pi}{6} - \left(-\cos \frac{\pi}{6} - \frac{1}{2} \cdot \frac{\pi}{6} \right) \right)$$

$$= 2 \left(\frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 \right) + \left(\frac{\sqrt{3}}{2} - \frac{5\pi}{12} + \frac{\sqrt{3}}{2} + \frac{\pi}{12} \right)$$

$$= 2\sqrt{3} - \frac{\pi}{6} - 2$$

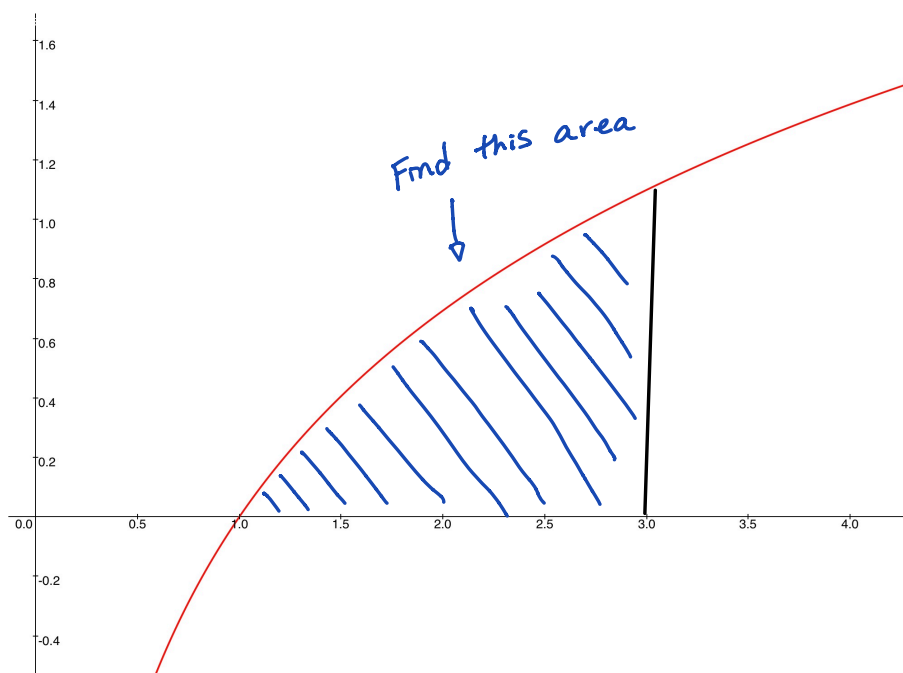
What if it is NOT a function of x ?



The area between $f(y)$ and $g(y)$ is

$$\int_c^d (\overset{\text{right}}{f(y)} - \overset{\text{left}}{g(y)}) dy$$

Ex (4) Find the area between
 $y = \ln x$ and the x -axis
on the interval $[1, 3]$

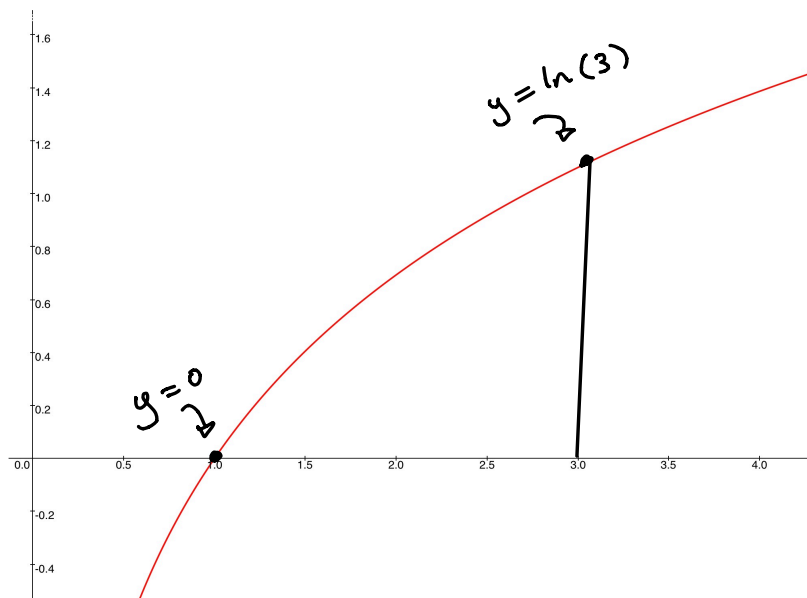


Notice that we cannot do $\int \ln x \, dx$ until Math 3B. So, change this to a function of y .

$$y = \ln x \rightarrow x = e^y$$

This is the function on the left.
The function on the right is $x=3$.

What are the limits of integration?



$$\text{Area} = \int_0^{\ln 3} [3 - e^y] dy$$

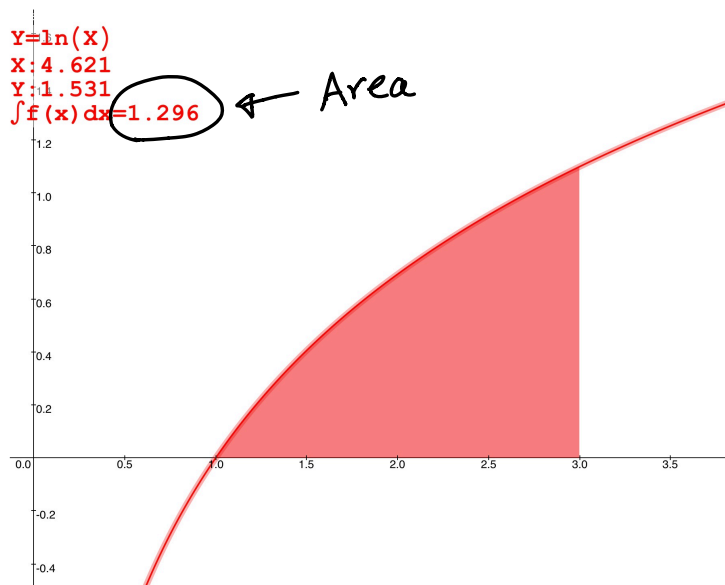
$$= (3y - e^y) \Big|_0^{\ln 3}$$

$$= 3 \cdot \ln 3 - e^{\ln 3} - (3(0) - e^0)$$

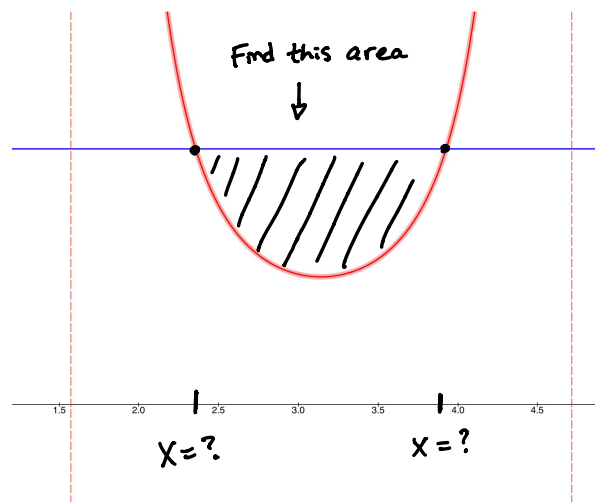
$$= 3 \cdot \ln 3 - 2$$

$$= 1.296$$

Now graph the original problem and check the answer.



Ex ⑤ Find the area between one cycle of $y = \sec^2 x$ and $y = 2$.

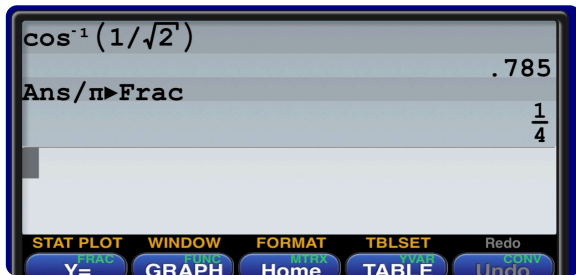


First we need to find the two intersection points. We will use these for the limits.

$$\sec^2 x = 2$$

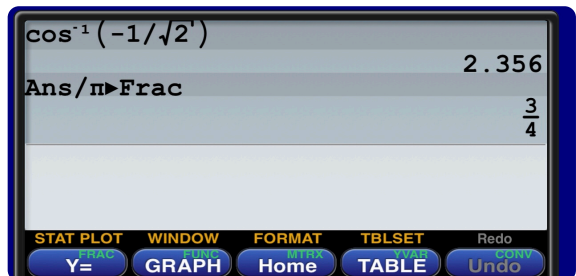
$$\frac{1}{\cos^2 x} = 2$$

$$\cos^2 x = \frac{1}{2}$$



$$\cos x = \pm \frac{1}{\sqrt{2}}$$

$$x = \frac{\pi}{4}$$



$$x = \frac{3\pi}{4}$$

$$\text{Area} = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\overset{\text{top} - \text{bottom}}{2 - \sec^2 x}) dx$$

$$= (2x - \tan x) \Big|_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= \left(2 \cdot \frac{5\pi}{4} - \tan \frac{5\pi}{4} \right) - \left(2 \cdot \frac{\pi}{4} - \tan \frac{\pi}{4} \right)$$

$$= \left(\frac{5\pi}{2} - 1 \right) - \left(\frac{\pi}{2} - 1 \right)$$

$$= 2\pi$$